5 A sphere of mass 3.0 kg travelling due North at 2.0 m s^{-1} collides with another sphere of mass 4.0 kg travelling due East at 2.0 m s^{-1}.

The magnitude of their resultant momentum after the collision will be

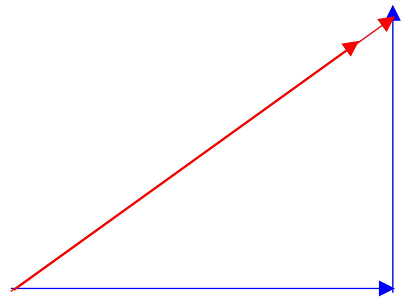
A 2.0 kg ms^{-1}.

B 10 kg m s^{-1}.

C 14 kg m s^{-1}.

D dependent on whether the the collision is elastic or inelastic.

Ans: B
Using vector addition, resultant initial momentum = 10 kg m s^{-1}
Since the final momentum = initial momentum (PCOM),
Magnitude of resultant momentum after collision is 10 kg m s^{-1}



6 A block and a sphere of equal mass m are placed on an inclined plane. If the maximum frictional